



Indian Oil developing batteries from indigenous metals to power EVs

The electric vehicle (EV) revolution in India is well underway, but one of the biggest hurdles is lack of charging infrastructure and reliance of Indian EV makers on batteries imported from China and other countries. To tackle both these deficiencies in one go, state-owned Indian Oil Corp. (IOC) is looking to build up India's capacity for developing indigenous batteries powered with new metals.

The move would prevent dependence of battery pack makers on lithium imports in the coming months, as well as reduce the burden on EV players to create robust charging infrastructure, according to a report in a leading English daily.

Sanjiv Singh, chairman of IOC, said that the company has big plans for the EV sector and green energy projects. He said, "Metal-air batteries will produce energy by oxidising such metals as iron, zinc and aluminium. These batteries cannot be

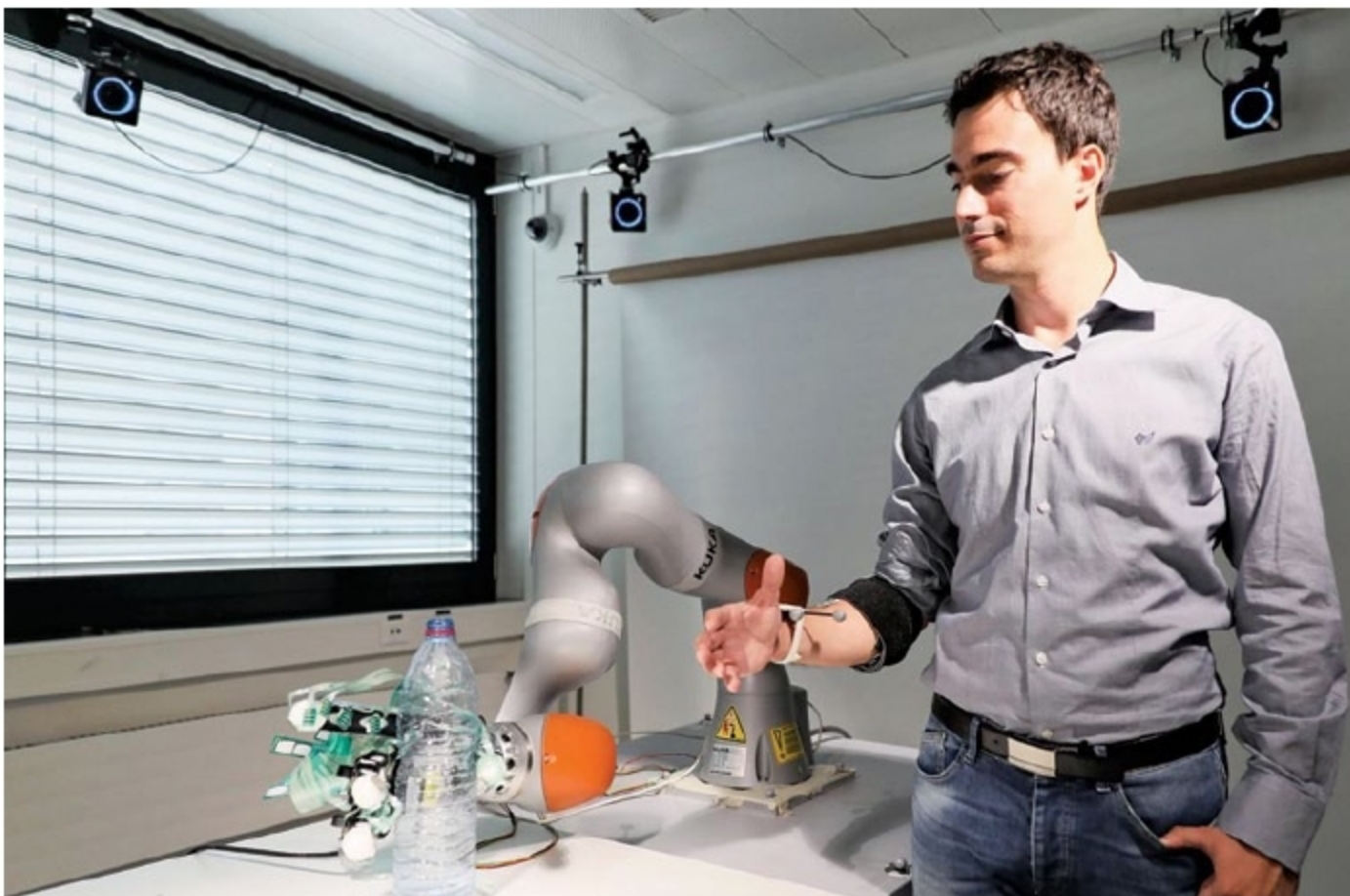
recharged, but one can replace plates to power them up almost instantly, making it much more convenient for users, thereby eliminating the need for charging infrastructure."

Singh added that, for a typical car user, the new system would be similar to going to the petrol pump for a refill. He said the company's pumps would be able to replace the metal plates that power up these batteries, and the whole process should only take three minutes.

Singh added, "Metal-air batteries have other advantages. These have high energy density. If a normal lithium-ion battery can travel the vehicle up to 300 kilometres, these indigenous metals powered batteries can take it to 500 kilometres."

According to IOC, metal oxide in the batteries can be reduced to metal again, making it a recyclable and reusable battery pack.

A smart artificial hand for amputees merges user and robotic control



Researcher Artoni Fiorenzo demonstrates the shared control system, using a robotic arm (Credit: Ecole Polytechnique Fédérale de Lausanne)

Scientists from Ecole Polytechnique Fédérale de Lausanne (EPFL) are developing new approaches for improved control of robotic hands—in particular, for amputees—that combine individual finger control and automation for improved grasping and manipulation. This interdisciplinary proof-of-concept between neuro-engineering and robotics was successfully

tested on three amputees and seven healthy subjects.

The technology merges two concepts from two different fields. Implementing them both together had never been done before for robotic hand control, and it contributes to the emerging field of shared control in neuro-prosthetics.

One concept, from neuro-engineering, involves deciphering intended finger movement from muscular activity on the amputee's stump for individual finger control of the prosthetic hand. The other, from robotics, allows the robotic hand to help take hold of objects and maintain contact with them for robust grasping.

"When you hold an object in your hand, and it starts to slip, you only have a couple of milliseconds to react," explains Aude Billard, who leads EPFL's Learning Algorithms and Systems Laboratory. "The robotic hand has the ability to react within 400 milliseconds. Equipped with pressure sensors all along the fingers, it can react and stabilise the object before the brain can actually perceive that the object is slipping."